

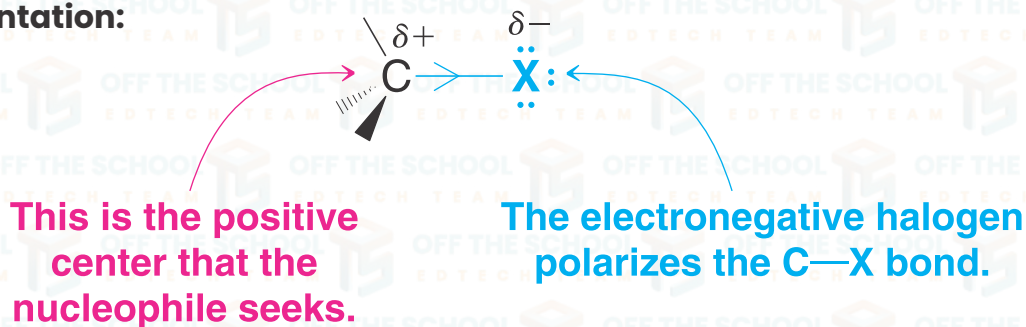
UNIT 06

ALKYL HALIDES & AMINES



Alkyl halides, also known as haloalkanes, are a group of chemical compounds in which one or more halogen atoms (fluorine, chlorine, bromine, or iodine) are bonded to an alkyl group (a saturated carbon chain).

Representation:



General Formula: The general formula for an alkyl halide depends on the type and number of halogens attached as well as the structure of the alkyl group. For a simple mono-halogenated alkane (where only one halogen atom is attached to an alkyl group), the general formula can be represented as $C_nH_{(2n+1)}X$.

Conversion to Alkyl Halides



ethyl fluoride, bp -38°C
smallest surface area, lowest bp



ethyl chloride, bp 12°C

Space-filling drawings of the ethyl halides. The heavier halogens are larger, with much greater surface areas. As a result, the boiling points of the ethyl halides increase in the order $F < Cl < Br < I$.



ethyl bromide, bp 38°C



ethyl iodide, bp 72°C
largest surface area, highest bp

Nucleophile:

A nucleophile is (an atom, ion, or molecule) that has an electron pair it can donate to form a new chemical bond. Nucleophiles are "nucleus-loving" because they are attracted to positive charges or electron-deficient sites.

Substrate:

The alkyl halide molecule on which a nucleophile attacks is called Substrate molecule.

Leaving Group:

A leaving group is also a nucleophile and an atom or group of atoms that detaches (with an unshared pair of electrons) from the rest of the molecule during a chemical reaction, particularly in substitution and elimination reactions.

Carbocation:

A carbocation is a positively charged ion (or species) that contains a carbon atom with only six electrons in its valence shell, making it electron-deficient. It is characterized by the presence of a trivalent carbon atom, making it highly reactive in organic reactions.

- Carbocations are reaction intermediates in following reactions:
- SN1 (unimolecular nucleophilic substitution).
- E1 reaction unimolecular elimination
- Electrophilic addition

Nucleophilic substitution reactions:

A class of chemical reactions characterized by the exchange of nucleophiles on a substrate. Where a strong nucleophile, replaces a weaker nucleophile, or leaving group, that is attached to an electrophilic carbon on the substrate. General Reaction Mechanism The basic process by which nucleophiles are exchanged in these reactions.



The nucleophile donates an electron pair to the substrate.

The bond between the carbon and the leaving group breaks, giving both electrons from the bond to the leaving group.

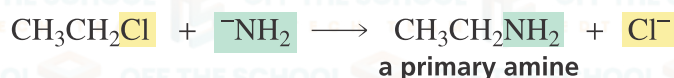
The nucleophile uses its electron pair to form a new covalent bond with the substrate carbon.

The leaving group gains the pair of electrons that originally bonded it in the substrate.

Common Reaction Example: How alkyl halides react with specific nucleophiles to form new products.

- **Reaction with Sodium Hydroxide:** Treating alkyl halides with NaOH to produce alcohols.
- **Reaction with Potassium Hydroxide:** Using KOH as a nucleophile in the substitution process.





Nucleophilic substitution reactions are categorized into two main mechanisms:

1. SN1 Reactions
2. SN2 Reactions

SN1 (Substitution Nucleophilic unimolecular) reaction:

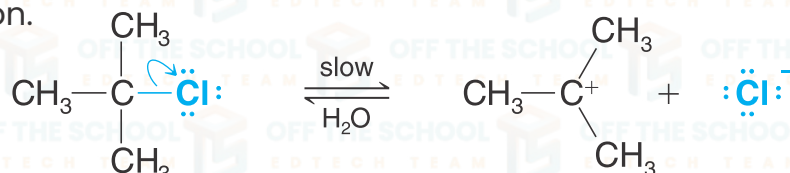
“The nucleophilic substitution reaction mechanism in which breaking of C–X bond and the formation of C–Nu bond one after the other in two different steps.”

Mechanism: This mechanism is unimolecular and consists of two distinct steps:

• Step 1: Formation of Carbocation:

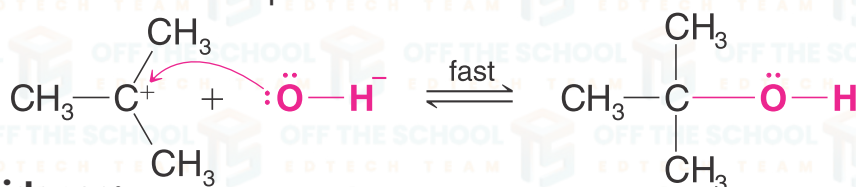
The reaction initiates with the departure of the leaving group (usually a halogen atom) from the substrate, resulting in the formation of a carbocation intermediate.

This is the rate-determining step and occurs slowly. For example, in tert-butyl bromide (a tertiary alkyl halide), the bromine atom leaves, forming a tertiary carbocation.



• Step 2: Nucleophilic Attack:

In the subsequent fast step, a strong nucleophile attacks the carbocation, leading to the formation of the final product.



Kinetic evidence:

$$R = k [\text{substrate}]$$

SN2 (Substitution Nucleophilic bimolecular) reaction:

“The nucleophilic substitution reaction mechanism in which breaking of C–X bond and the formation of C–Nu bond one after the other in one step.”

Mechanism: This mechanism is unimolecular and consists of one step:

1. Backside Attack: In SN2 reactions, the nucleophile attacks the carbon from the opposite side of the leaving group.

2. Steric Hindrance: SN2 reactions are sensitive to crowding around the carbon atom. The less crowded (or less steric hindrance), the faster the reaction will be.

3. Leaving Group: A leaving group is a part of the original molecule that gets replaced. It must be able to leave easily, meaning it can comfortably take on a pair of electrons.



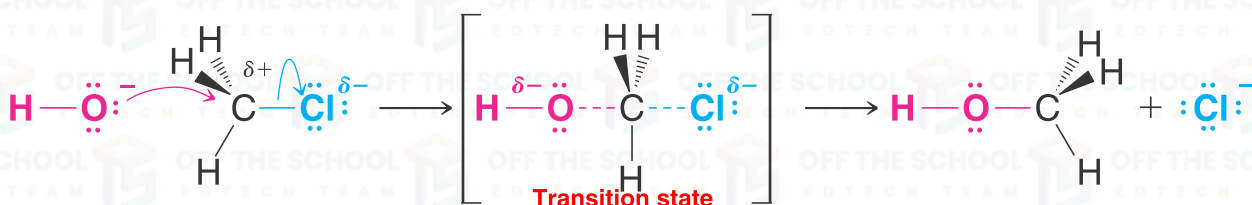
4. Rate of Reaction: The rate depends on the concentration of both the nucleophile and the substrate (the molecule being attacked).

5. One Step: The entire S_N2 reaction happens in one concerted step, without intermediates. This is why it's known as a "concerted" reaction mechanism.

REACTION



MECHANISM



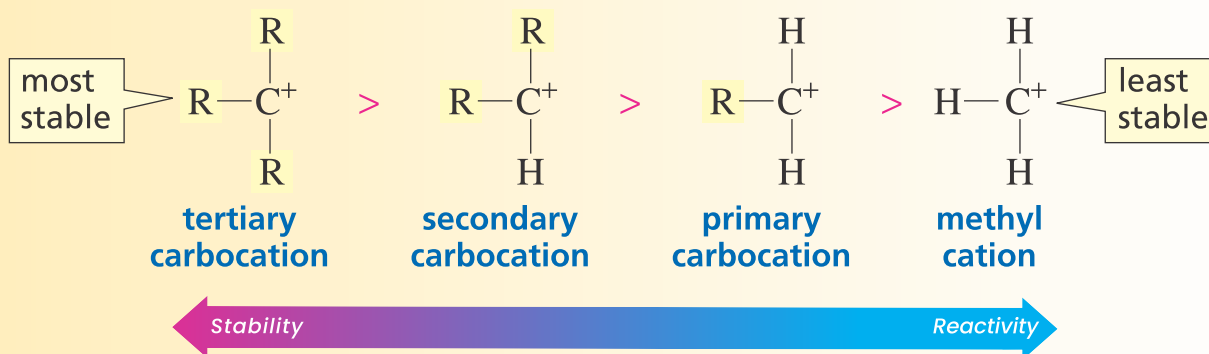
Kinetic evidence:

$$R = k [\text{substrate}] [\text{Nu}^-]$$

What is A Carbocation?

A positively charged carbon atom bearing three covalent bonds and an empty orbital is called a carbocation (or more officially, a "carbenium" ion)

relative stabilities of carbocations

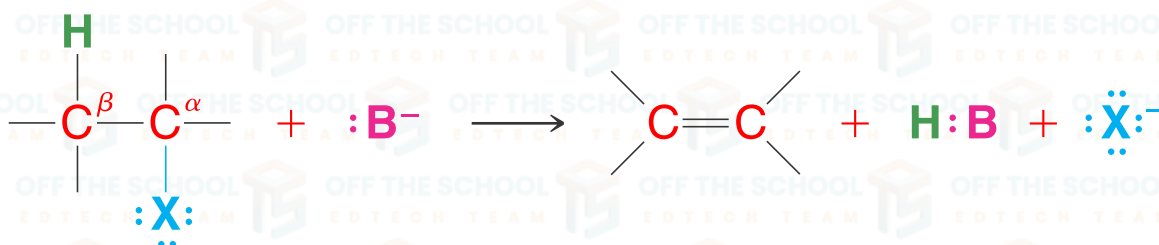


Elimination Reaction:

"An elimination reaction is a type of organic reaction where two atoms or groups are removed from a molecule, resulting in the formation of a double bond or a triple bond. This process typically produces an alkene or alkyne."

General characteristics of elimination reactions include:

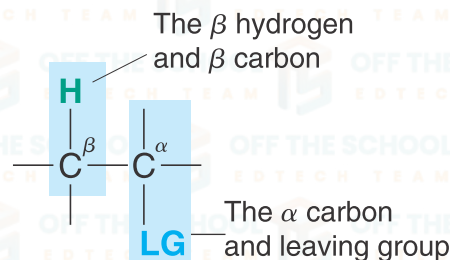
- 1. Formation of Multiple Bonds:** Elimination reactions often result in the formation of pi bonds, leading to alkenes or alkynes from single-bonded carbon atoms.
- 2. Leaving Groups:** A good leaving group is required, which is typically a halide that can depart with a pair of electrons.
- 3. Dehydrohalogenation:** In cases where a hydrogen atom and a halogen are removed, the reaction is specifically called dehydrohalogenation.



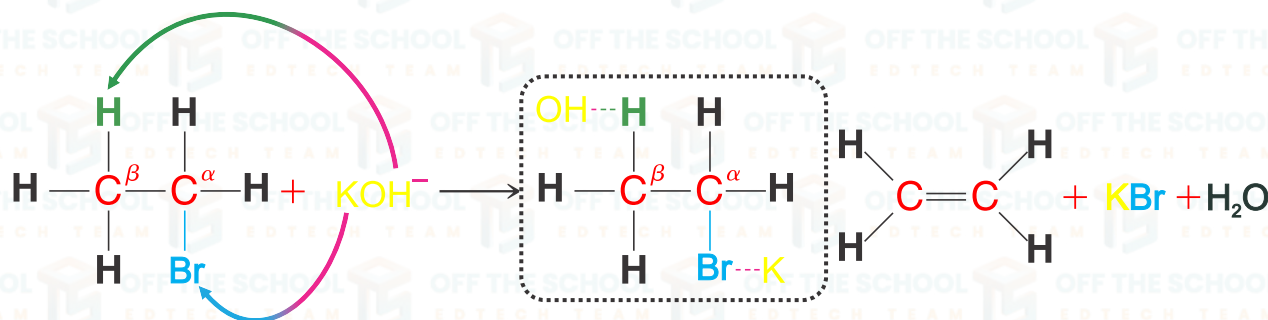
Classification of Elimination reaction:

The elimination reaction is classified as follows:

1. Bimolecular elimination reaction or E2 reaction.
2. Unimolecular elimination reaction E1 reaction.

**Bimolecular elimination reaction or E2 reaction:**

"A bimolecular elimination reaction, or E2 reaction, is a chemical process where two atoms or groups of atoms are removed from a molecule, forming a double bond. It occurs in a single step, with both the leaving group and the hydrogen atom being removed simultaneously."

**E2 Mechanism Overview**

The E2 mechanism is a one-step process where a base (B^-) removes a hydrogen atom (H) while the leaving group (X^-) departs, forming a double bond, resulting in an alkene.

Base and Hydrogen Interaction:

A strong base (B^-) attracts and bonds with a hydrogen atom (H) attached to the carbon next to the carbon bonded to the leaving group (X^-).

Formation of Transition State:

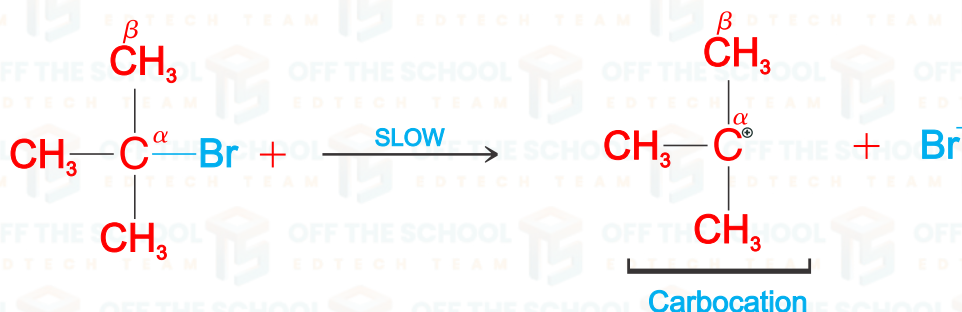
As the base pulls the hydrogen away, electrons move towards forming a new double bond, creating a transition state with a partial double bond.

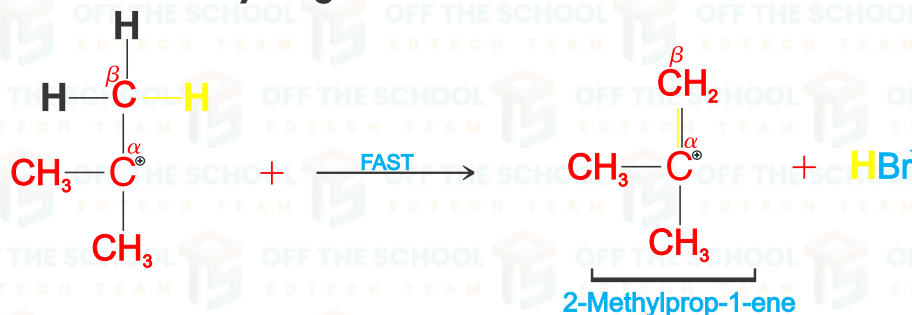
Alkene Creation:**Example: Ethene Formation:**

When hydroxide (HO^-) acts as the base with ethyl bromide ($\text{CH}_3\text{CH}_2\text{Br}$), it removes a hydrogen from the ethyl group, and bromide (Br^-) leaves, forming ethene (C_2H_4), water (H_2O), and bromide ion (Br^-).

Unimolecular elimination reaction E1 reaction:

"A unimolecular elimination reaction, or E1 reaction, is a two-step chemical process where a molecule loses a group of atoms (usually a halogen) to form a carbocation intermediate first. Then, a hydrogen atom is removed, usually by a weak base, leading to the formation of a double bond in the resulting alkene."

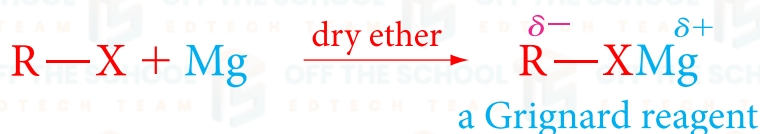
STEP - 01: CARBOCATION

STEP-02: Removal of Hydrogen atom from beta-Carbon**GRIGNARD'S REAGENT**

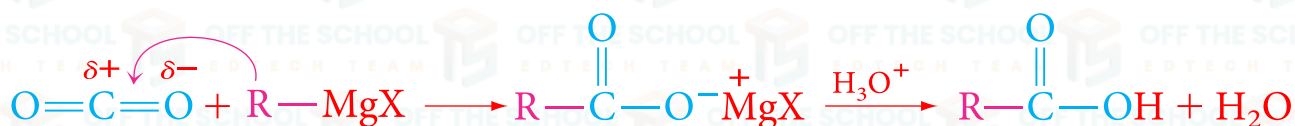
- François Auguste Victor Grignard (1871-1935) was a Nobel Prize winning French chemist.
- In 1900, V. Grignard reported that an alkyl halide (RX) reacts with magnesium metal (Mg) in diethyl ether to give a cloudy solution of an organomagnesium compound (RMgX).
- He was awarded the Nobel Prize in Chemistry in 1912.
- The Grignard reagent is usually described with the general chemical formula;



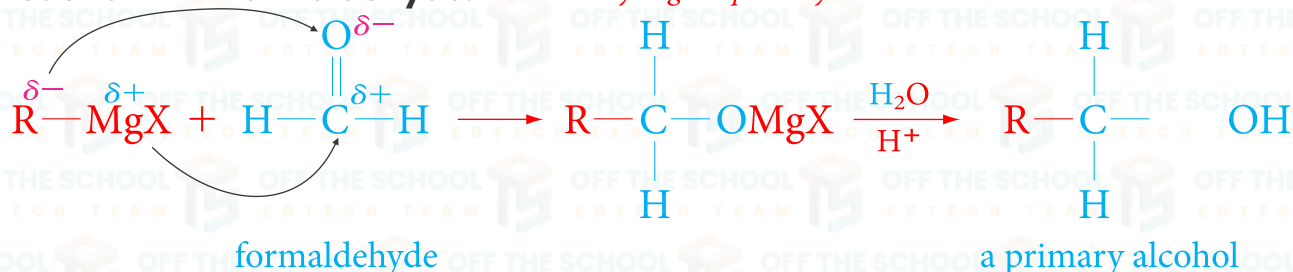
- Iodides, bromides, and chlorides can be used, including aryl and alkyl halides.

**Reactivity:**

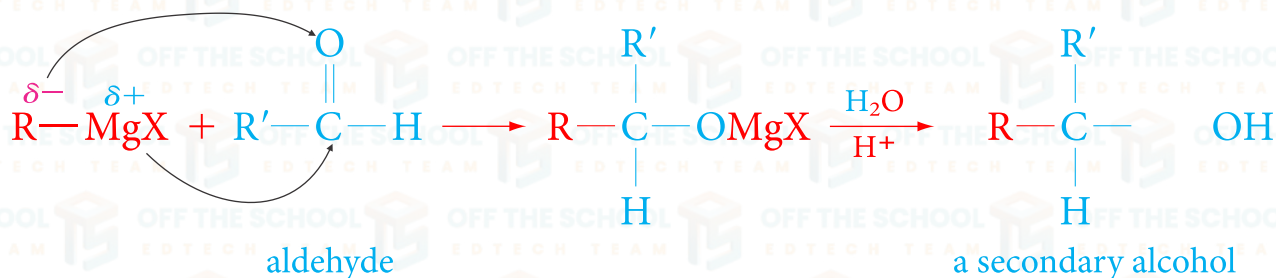
In Grignard reagents, the bond between carbon and magnesium is polar because carbon is more electronegative than magnesium. This makes the carbon part rich in electrons and gives it a negative charge, so it acts as a nucleophile. Because of this, Grignard reagents can react with molecules that have a positive part by either substituting (replacing) or adding to them, helping to build larger organic molecules.

Reaction with Carbon dioxide:

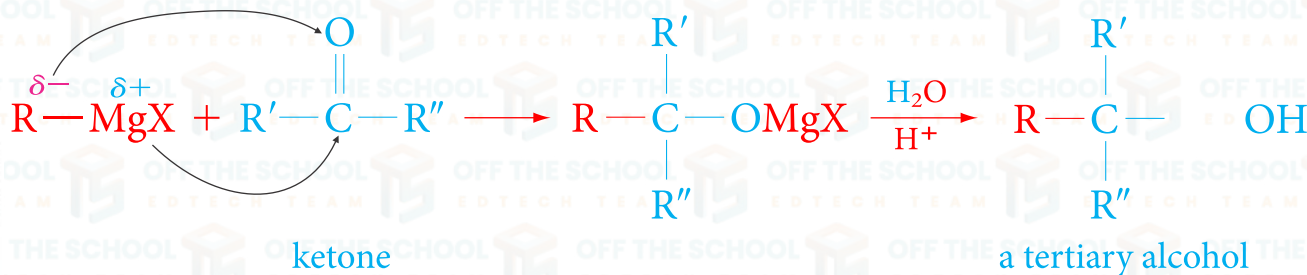
Reaction with Formaldehyde: *Formaldehyde gives primary alcohols.*

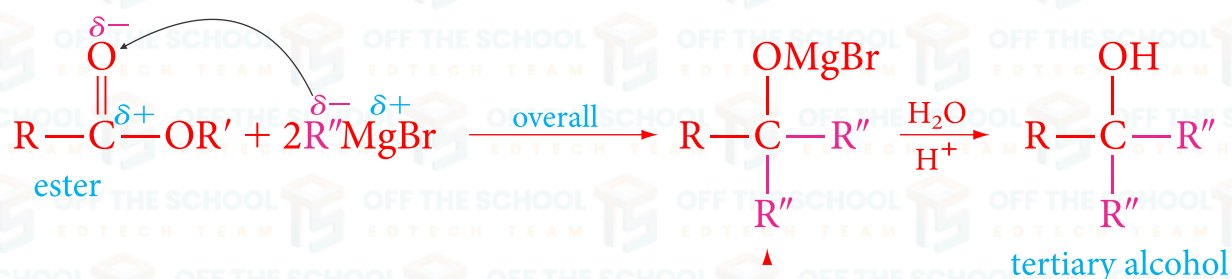
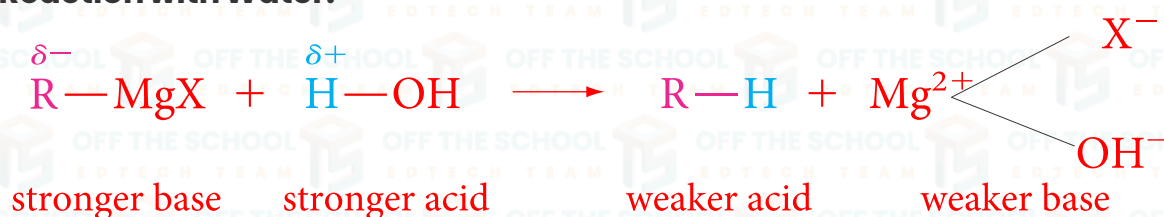


Reaction with Acetaldehyde: *Other aldehydes give secondary alcohols.*



Reaction with Acetone: *Ketones give tertiary alcohols.*

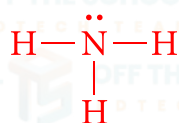


Reaction with Ester:**Reaction with Water:**

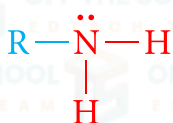
Amines: Amines are chemicals made from ammonia by replacing one or more hydrogen atoms with carbon chains or rings. (NH_2^-)

Classification and Structure of Amines

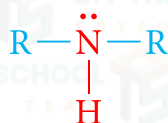
The relation between ammonia and amines is illustrated by the following structures:



ammonia



primary amine



secondary amine



tertiary amine

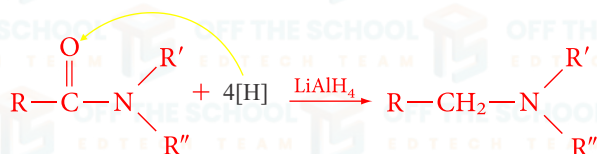
REACTION SUMMARY OF AMINES

1. Alkylation of Ammonia and Amines to Form Amines



2. Reduction Routes to Amines

a. Hydride Reduction of Amides and Nitriles



b. Reductive Amination of Aldehydes and Ketones



3. Amines as Bases



4. Amines as Nucleophiles

a. Acylation of Amines

Secondary and Tertiary Amides from Primary and Secondary Amines

